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Module Code	**CCS3440**
Module Name	Artificial Intelligence
Credit Value	4 Credits
Assessment Type	Coursework
Total Marks	100
Submission Type	Group (Maximum 4 Students)
Dataset	SmartCare Hospital AI Dataset
Course objectives	This coursework aims to enable students to: • Understand the complete AI project lifecycle. • Apply machine learning techniques to healthcare problems. • Perform data preprocessing and feature engineering. • Conduct exploratory data analysis. • Develop predictive models. • Compare multiple AI algorithms. • Evaluate model performance. • Interpret predictions using Explainable AI techniques. • Develop a simple AI-powered decision support prototype.

Intended Learning Outcome (ILO) Alignment

ILO	Description	Weight
ILO1	Explain fundamental AI concepts and machine learning techniques.	15%
ILO2	Analyze and formulate real-world problems as AI solutions.	20%
ILO3	Develop and implement machine learning models using appropriate tools and algorithms.	30%
ILO4	Evaluate AI solutions using performance metrics and best practices.	20%
ILO5	Communicate and justify AI solutions through reports, presentations, and demonstrations.	15%

Background

Healthcare organizations generate large volumes of patient, clinical, operational, and financial data. Extracting meaningful insights from this data is critical for improving patient care, reducing costs, and supporting evidence-based decision-making.

SmartCare Hospital has accumulated a large dataset containing patient demographics, clinical measurements, appointment history, admission records, treatment information, laboratory tests, and billing information.

Hospital management wishes to utilize Artificial Intelligence (AI) and Machine Learning (ML) techniques to develop predictive models that can assist in operational planning and clinical decision support.

Students are required to design, develop, evaluate, and interpret AI models using the provided SmartCare Hospital dataset.

Dataset Provided

Students will be provided with:

Dataset File

smartcare_ai_dataset_1000.csv

Data Dictionary

smartcare_ai_dataset_data_dictionary.csv

The dataset contains 1000 hospital records and has been specifically designed for Artificial Intelligence and Machine Learning coursework.

Students must use the provided dataset.

No external datasets are required.

Dataset Description

The dataset contains information related to:

Patient Information

- Patient ID
- Age
- Gender
- Blood Group

Clinical Information

- Diagnosis
- Blood Pressure
- Blood Sugar
- Cholesterol
- BMI

Hospital Operations Data

- Department
- Appointment History
- Previous Admissions
- Length of Stay
- Room Type
- Treatment Count
- Laboratory Test Count

Financial Data

- Consultation Charges
- Laboratory Charges
- Room Charges

- Medicine Charges
- Total Bill Amount

Available AI Prediction Problems

Students must select ONE of the following prediction tasks.

Option A – Appointment No-Show Prediction

Problem Statement

Many patients fail to attend scheduled appointments, resulting in inefficient resource utilization and lost revenue.

Develop an AI model to predict whether a patient is likely to miss an appointment.

Target Variable

no_show

Problem Type

Binary Classification

Output

- No Show
- Attended

Option B – Patient Readmission Prediction

Problem Statement

Patients who are re-admitted shortly after discharge can increase healthcare costs and indicate ineffective treatment planning.

Develop an AI model to predict whether a patient will be re-admitted within 30 days.

Target Variable

readmitted_30_days

Problem Type

Binary Classification

Output

- Readmitted
- Not Readmitted

Option C – Disease Risk Classification

Problem Statement

Early identification of disease risk can improve preventive healthcare interventions.

Develop an AI model to classify patients into disease risk categories.

Target Variable

disease_risk_level

Problem Type

Multi-Class Classification

Classes

- Low
- Medium
- High

Technologies to be Used

Students must use:

Programming Language

Python

Development Environment

Jupyter Notebook

Data Analysis Libraries

- Pandas
- NumPy

Visualization Libraries

- Matplotlib
- Seaborn

Machine Learning Libraries

- Scikit-Learn

Optional:

- XGBoost
- TensorFlow
- Keras

Explainable AI Libraries

- SHAP
- LIME

Prototype Development

One of the following:

- Streamlit
- Flask

Coursework Tasks

Task 01 – Problem Definition and Literature Review

Requirements

Students must:

- Define the selected AI problem.
- Explain its significance.
- Conduct a literature review.

Minimum Requirements:

- Five peer-reviewed research papers.

Students should discuss:

- Existing AI approaches.
- Common algorithms.
- Current limitations.
- Research gaps.

Deliverables

- Problem Definition
- Literature Review
- Research Gap Analysis

Marks

10 Marks

Task 02 – Dataset Understanding

Requirements

Students must:

- Explore the provided dataset.
- Identify input and target variables.
- Explain attribute meanings.
- Discuss data quality.

Deliverables

- Dataset Overview
- Attribute Description
- Data Dictionary Interpretation

Marks

10 Marks

Task 03 – Data Preprocessing and Feature Engineering

Requirements

Perform:

- Missing value handling
- Duplicate record detection
- Outlier identification
- Data cleaning
- Feature encoding
- Feature scaling
- Feature selection
- Feature engineering

Students must justify all preprocessing decisions.

Deliverables

- Data Cleaning Workflow
- Feature Engineering Process

Marks

15 Marks

Task 04 – Exploratory Data Analysis (EDA)

Requirements

Perform:

- Descriptive statistics

- Correlation analysis
- Distribution analysis
- Pattern discovery

Required visualizations:

- Histograms
- Boxplots
- Scatterplots
- Correlation Heatmaps
- Class Distribution Charts

Students must discuss insights gained.

Deliverables

- Visualizations
- Statistical Analysis
- Interpretation

Marks

10 Marks

Task 05 – Machine Learning Model Development

Requirements

Students must train at least THREE machine learning models.

Examples include:

Classification Models

- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine
- Naïve Bayes
- K-Nearest Neighbors
- XGBoost

Students should compare model performance.

Deliverables

- Model Training
- Hyperparameter Selection
- Comparative Analysis

Marks

20 Marks

Task 06 – Model Evaluation

Requirements

Evaluate all models.

Binary Classification Metrics

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

Multi-Class Metrics

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

Students must identify and justify the best-performing model.

Deliverables

- Evaluation Results
- Model Comparison Table

Marks

10 Marks

Task 07 – Explainable AI Analysis

Requirements

Students must interpret model predictions using:

- SHAP OR
- LIME OR
- Feature Importance Analysis

Students must discuss:

- Important features
- Prediction reasoning
- Transparency
- Ethical implications

Deliverables

- Explainability Visualizations
- Interpretation Report

Marks

10 Marks

Task 08 – AI Prototype Development

Requirements

Develop a simple AI-powered prototype.

The prototype should:

- Accept user input.
- Process patient information.
- Generate predictions.
- Display results.

Students may use:

- Streamlit OR
- Flask

Deliverables

- Working Prototype
- Screenshots

Marks

5 Marks

Task 09 – Technical Report

Requirements

Prepare a professional report containing:

1. Introduction
2. Problem Definition
3. Literature Review
4. Dataset Description
5. Data Preprocessing
6. Exploratory Data Analysis
7. Feature Engineering
8. Model Development
9. Model Evaluation
10. Explainable AI Analysis
11. Prototype Design
12. Results and Discussion
13. Conclusion
14. References

Marks

5 Marks

Viva Examination

Students must individually explain:

- Dataset attributes
- Feature engineering decisions
- AI algorithms used
- Model comparison

- Evaluation metrics
- Explainable AI outputs
- Prototype functionality
- Individual contribution

Marks

5 Marks

Submission Requirements

Students must submit:

1. Technical Report (PDF)
2. Jupyter Notebook (.ipynb)
3. Python Source Code (.py)
4. Trained Model Files (.pkl / .joblib)
5. Prototype Source Code
6. GitHub Repository Link
7. Presentation Slides
8. Video Demonstration (5–10 minutes)

Assessment Criteria

Component	**Marks**
Problem Definition & Literature Review	10
Dataset Understanding	10
Data Preprocessing & Feature Engineering	15
Exploratory Data Analysis	10
Machine Learning Model Development	20
Model Evaluation	10
Explainable AI Analysis	10
AI Prototype Development	5
Technical Report	5
Viva Examination	5
Total	100

Bonus Marks (Up to 5 Marks)

Additional marks may be awarded for:

- Deep Learning Models
- Hyperparameter Optimization
- Ensemble Learning
- Advanced Explainable AI Techniques
- Cloud Deployment
- Multiple Prediction Tasks

Bonus marks are awarded at the lecturer's discretion.

Academic Integrity

Students must develop their own models, analyses, and reports.

AI tools such as ChatGPT, Gemini, Claude, GitHub Copilot, and similar systems may be used as learning aids; however, students must fully understand and be able to explain all code, preprocessing steps, algorithms, evaluation metrics, and outputs during the viva examination.

Any plagiarism, fabricated results, copied code, or academic misconduct will be handled according to university academic integrity regulations.

End of Coursework Specification